

PROBLEM STATEMENT: SIH26068

THEME: DISASTER MANAGEMENT & AI

WeatherGPT

Conversational AI for Weather Forecasting, IMD Alerts & Climate Information

Team Information

Team Name: Team WeatherGPT

Team Lead: Dev Vrajawat

Hackathon: Internal Hackathon 2026

Core Focus

Natural Language AI Weather Assistant,
IMD-style Alert Engine, and District
Climate Intelligence.

Project Links

Live Web App:

devvrajawat0.github.io/weathergpt

GitHub Repo:

github.com/devvrajawat0/weather-gpt-2

The Core Concept

WeatherGPT is a full-stack, chat-first climate assistant enabling users to ask complex natural-language questions ("Will it rain in Bhopal tomorrow?", "Compare weather in Delhi and Tokyo") and receive conversational, data-backed answers alongside interactive visual dashboards.

Comprehensive Data Coverage

- **700+ Indian Districts:** Grouped by State/UT with exact lat/long coordinates.
- **195+ World Capitals:** Complete global coverage for instant search.
- **Live Weather Metrics:** Temp, Feels-like, Wind, Humidity, Pressure, UV Index, Sunrise/Sunset, and 7-day trend.

Severe Weather & Alert Panel

Integrates an **IMD-style Alert Engine** evaluating live weather data against official warning thresholds for Heatwaves, Heavy Rainfall/Floods, Squall/Cyclones, and Severe AQI with color-coded severity:

GREEN Normal	YELLOW Be Updated	ORANGE Be Prepared	RED Take Action
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Conversational Voice & Multimodal Access

- **Voice Input (Web Speech API):** Speech-to-Text capability for hands-free query input.
- **Text-to-Speech (TTS):** Reads natural-language weather advisories aloud.
- **Smart Advisories:** Tailored clothing, outdoor travel, and farming advice.

⚙️ Full-Stack Technology Stack

React 18

TypeScript

Vite

Tailwind CSS

Node.js

Express

TanStack Query

Leaflet Maps

Anthropic SDK

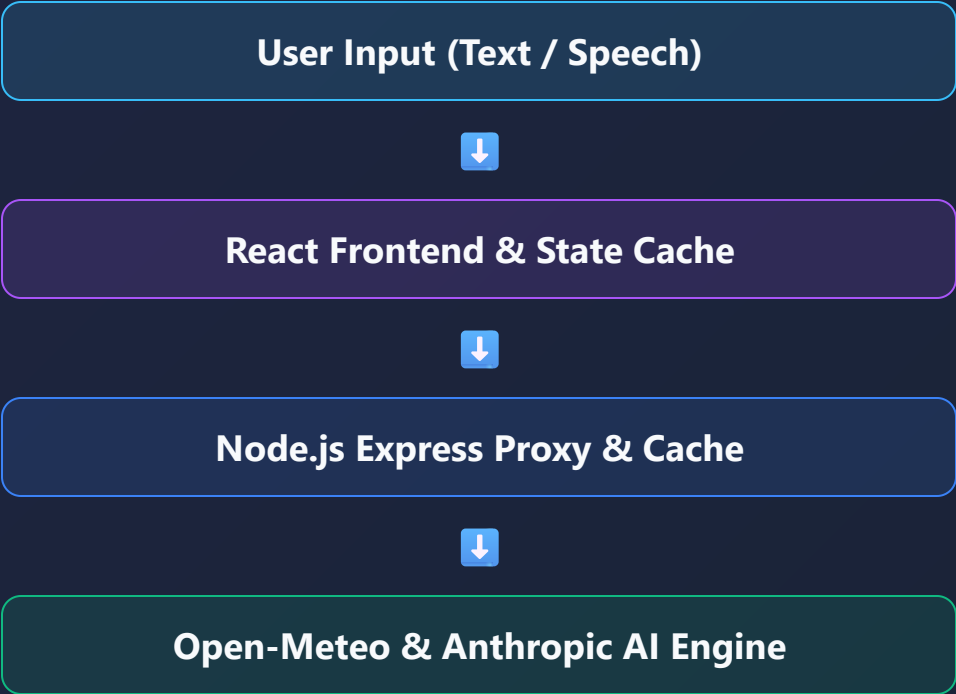
Open-Meteo API

- ➔ **Frontend Layer:** Component-based React UI with dynamic weather gradients, responsive breakpoints, dark mode, and Leaflet interactive map rendering.
- ➔ **Backend API Layer:** Node.js Express server handling rate limiting (`express-rate-limit`), CORS security, and 15-minute response caching (`node-cache`).
- ➔ **AI & Fallback Layer:** Anthropic Claude API proxy with live context injection + built-in Intelligent Fallback AI engine ensuring 100% offline client resilience.

📡 Data Sources & API Integration

- ➔ **Open-Meteo Forecast API:** Live 7-day daily & 24-hour hourly meteorological parameters.
- ➔ **Open-Meteo Air Quality API:** Real-time US AQI, PM2.5, PM10, NO2, O3, SO2 values.

🔄 Architecture Flowchart



💰 Zero-Cost Geocoding Architecture

Paid geocoding APIs (e.g. Google Maps API) incur high operational costs and rate limits. WeatherGPT solves this by pre-bundling a complete static dataset of **700+ Indian Districts** and **195+ World Capitals** with coordinate matrices, achieving **zero API cost & zero geocoding latency**.

⚡ High Availability & Double-Layer Caching

- ➔ **Backend Server Cache:** `node-cache` (15-min TTL) reduces external weather API calls by over 90%.
- ➔ **Frontend Query Cache:** TanStack React Query prevents re-fetching during tab switching.
- ➔ **API Quota Efficiency:** Open-Meteo free tier provides 10,000 requests/day per IP, ensuring extreme financial viability.

🛡️ Offline & Static Resilience

WeatherGPT features an intelligent client-side fallback system. If the Express backend server is unreachable (e.g., when hosted on static platforms like GitHub Pages), the app automatically switches to direct browser API fetching and client-side AI synthesis with **100% uptime guarantee**.

📈 Scalability Matrix

Dimension	Strategy
API Rate Limits	15-minute TTL caching & Open-Meteo free tier
Deployment	Docker-ready monorepo (Vercel / Surge / GitHub Pages)
Security	API key proxying, input sanitization & rate limiting

IMPACT & BENEFITS

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Agricultural Weather Support for Farmers

Provides tailored farming advisories covering irrigation scheduling, pesticide application windows based on wind speed, and crop harvest protection ahead of predicted rain showers.

Disaster Management & Emergency Preparedness

Delivers early warnings for extreme heatwaves, flash floods, squalls, and air pollution spikes with actionable safety instructions and emergency helpline contacts (NDRF: 1078, IMD: 1800-180-1717).

Accessibility & Digital Inclusion

Web Speech API voice input and text-to-speech reading allow rural populations and non-literate users to access critical weather and climate advisories effortlessly.

Public Health & AQI Awareness

Monitors real-time Air Quality Index (US AQI) and fine particulate matter (PM2.5/PM10), warning vulnerable populations to take protective health precautions.

RESEARCH & REFERENCES

Standards & Scientific Frameworks

- **IMD Alert Criteria:** India Meteorological Department warning thresholds for Heatwaves ($\geq 40^{\circ}\text{C}$), Heavy Rainfall ($\geq 64.5\text{mm}$), and Squalls.
- **CPCB Air Quality Index:** Central Pollution Control Board standards for US AQI, PM2.5, PM10, and gaseous pollutants.
- **WMO Weather Codes:** World Meteorological Organization code mapping for international weather condition standardization.

Meteorological API Documentation

- Open-Meteo Weather Forecast API: `open-meteo.com/en/docs`
- Open-Meteo Air Quality API: `open-meteo.com/en/docs/air-quality-api`

Project Deliverables & Submission Links

Deliverable	URL / Location
Live Web App	<code>devvrajawat0.github.io/weathergpt/</code>
GitHub Repository	<code>github.com/devvrajawat0/weather-gpt-2</code>
Alternative Mirror	<code>weathergpt2.surge.sh</code>

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